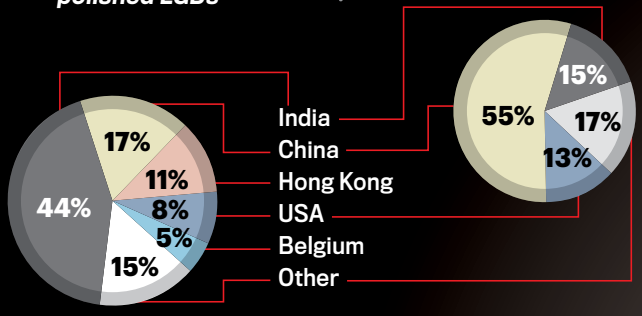
A CUT ABOVE

The processes involved in producing lab diamonds, India's growing role in their production and export, and their uses

LEADING ROLE

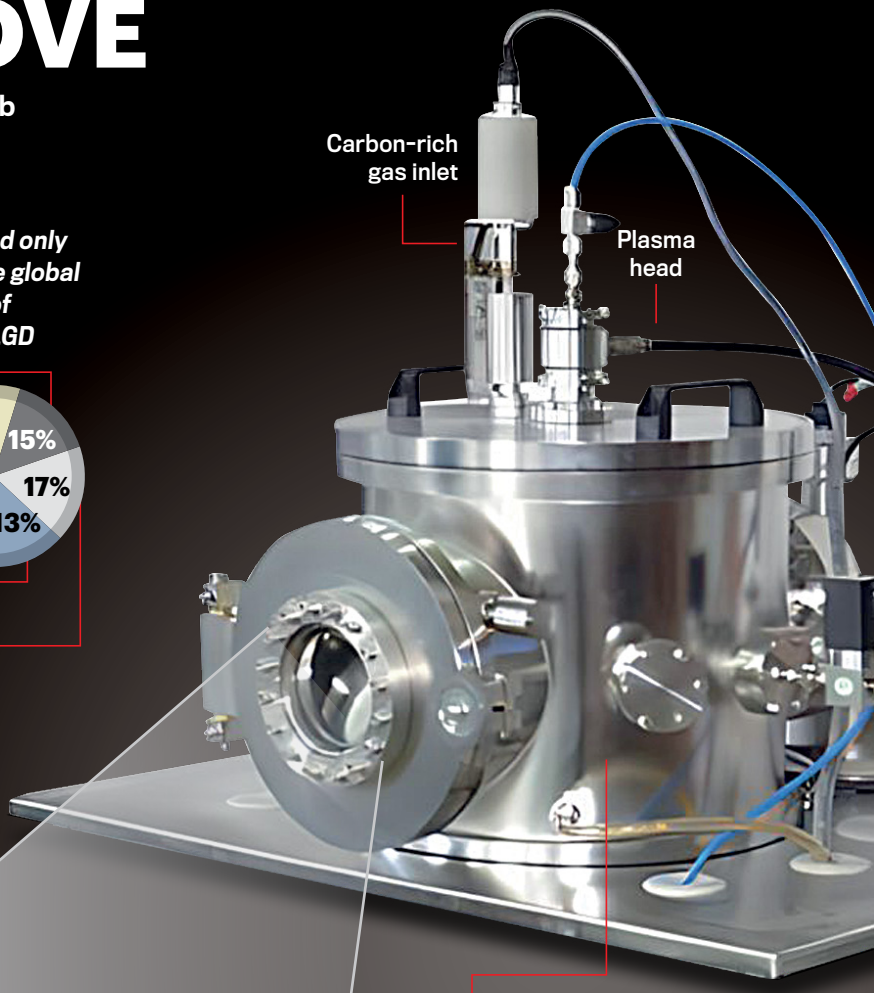
*India leads the global **exports** of cut and polished LGDs*

*India is second only to China in the global **production** of gem-quality LGD*



$15 BILLION
Projected LGD market in India by 2035

2.5 MILLION
Projected jobs in India's LGD sector by 2025





CULTIVATING A DIAMOND

CVD PROCESS



A thin diamond segment, called the seed crystal, is placed in a sealed chamber

The chamber is heated to around 800°C and flooded with carbon-rich gas. The gas ionises, breaking down into pure carbon



The pure carbon molecules attach to the original diamond seed, layer by layer



This process continues until a fully formed rough diamond is created



It is then cut and polished using the traditional methods employed for natural diamonds

Graphic by **TANMOY CHAKRABORTY**